Trusting Oracle Integration in the Fusion Identity Domain and Proving a JWT User Assertion Token

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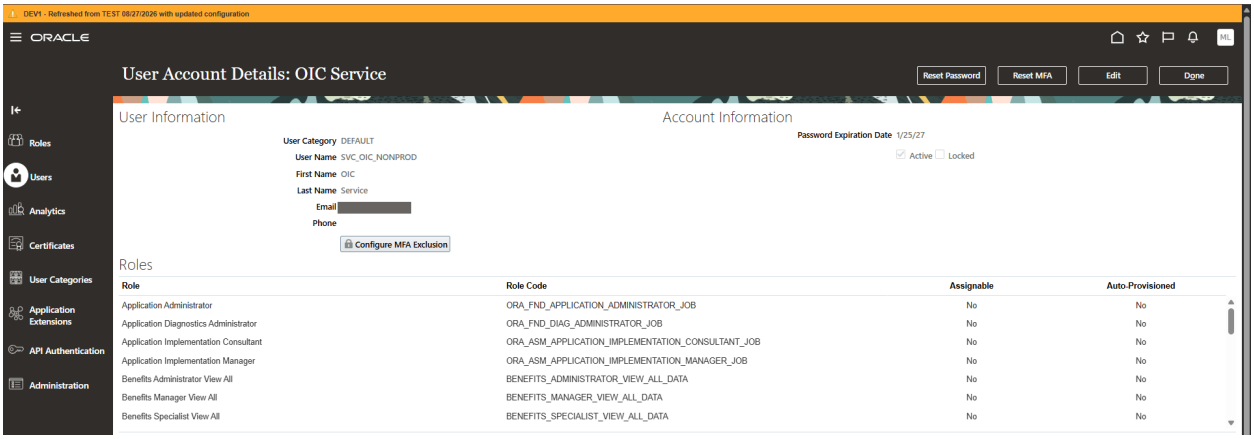
Product: OCI IAM Identity Domains, Oracle Fusion Cloud Applications, Oracle Integration 3

Note: environment identifiers (OCIDs, client IDs, identity domain and pod identifiers, and email addresses) are masked in the screenshots.

With OAuth using JWT User Assertion, Oracle Integration asserts the identity of a Fusion integration user by signing a token request with its private key. The identity domain that issues the token is the one attached to the Fusion environment, not Oracle Integration's own domain. This contribution shows how I set up that trust and proved a token from Cloud Shell before changing any live Oracle Integration connection.

Part 1: Confirm the Fusion integration user

The user named in the sub claim of the assertion must exist and be active in the Fusion identity domain. I opened the integration user in the Fusion Security Console and confirmed it was active. I did not add any roles for OAuth: the user keeps the same roles it used with Basic Authentication. I also noted the password expiry date, because the password still matters until every connection is switched.



The Fusion integration user, active, with its password expiry date.

Roles

Role	Role Code	Assignable	Auto-Provisioned
Benefits Specialist View All	BENEFITS_SPECIALIST_VIEW_ALL_DATA	No	No
Compensation Administrator View All	COMPENSATION_ADMINISTRATOR_VIEW_ALL_DATA	No	No
Compensation Analyst View All	COMPENSATION_ANALYST_VIEW_ALL_DATA	No	No
Compensation Manager View All	COMPENSATION_MANAGER_VIEW_ALL_DATA	No	No
Employee	ORA_PER_EMPLOYEE_ABSTRACT	No	No
General Accountant	ORA_GL_GENERAL_ACCOUNTANT_JOB	No	No
General Accounting Manager	ORA_GL_GENERAL_ACCOUNTING_MANAGER_JOB	No	No
Hiring Manager	ORA_IRC_HIRING_MANAGER_ABSTRACT	No	No

Roles

Role	Role Code	Assignable	Auto-Provisioned
Hiring Manager	ORA_IRC_HIRING_MANAGER_ABSTRACT	No	No
Human Capital Management Integration Specialist	ORA_HRC_HUMAN_CAPITAL_MANAGEMENT_INTEGRATION_SPECIALIST_JOB	No	No
Human Resource Analyst View All	HUMAN_RESOURCE_ANALYST_VIEW_ALL_DATA	No	No
Human Resource Manager View All	HUMAN_RESOURCE_MANAGER_VIEW_ALL_DATA	No	No
Human Resource Specialist	ORA_PER_HUMAN_RESOURCE_SPECIALIST_JOB	No	No
Human Resource Specialist View All	HUMAN_RESOURCE_SPECIALIST_VIEW_ALL_DATA	No	No
IT Security Manager	ORA_FND_IT_SECURITY_MANAGER_JOB	No	No
Integration Specialist	ORA_FND_INTEGRATION_SPECIALIST_JOB	No	No

Roles

Role	Role Code	Assignable	Auto-Provisioned
Integration Specialist	ORA_FND_INTEGRATION_SPECIALIST_JOB	No	No
Line Manager	ORA_PER_LINE_MANAGER_ABSTRACT	No	No
Payroll Administrator View All	PAYROLL_ADMINISTRATOR_VIEW_ALL_DATA	No	No
Payroll Manager View All	PAYROLL_MANAGER_VIEW_ALL_DATA	No	No
Recruiting Administrator	ORA_PER_RECRUITING_ADMINISTRATOR_JOB	No	No
Recruiting Administrator View All	RECRUITING_ADMINISTRATOR_VIEW_ALL_DATA	No	No
Recruiting Manager View All	RECRUITING_MANAGER_VIEW_ALL_DATA	No	No
Time and Labor Administrator View All	TIME_AND_LABOR_ADMINISTRATOR_VIEW_ALL_DATA	No	No

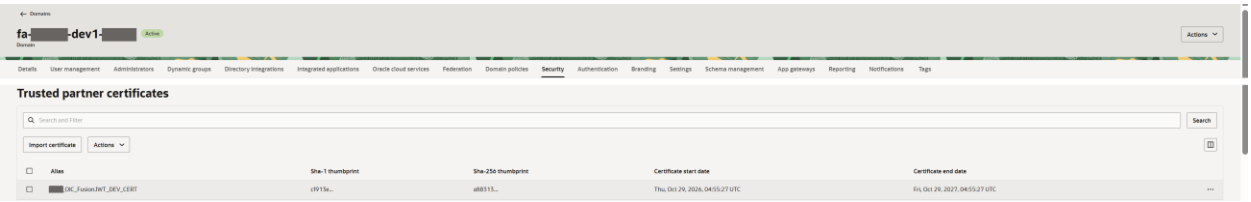
Roles

Role	Role Code	Assignable	Auto-Provisioned
Payroll Administrator View All	PAYROLL_ADMINISTRATOR_VIEW_ALL_DATA	No	No
Payroll Manager View All	PAYROLL_MANAGER_VIEW_ALL_DATA	No	No
Recruiting Administrator	ORA_PER_RECRUITING_ADMINISTRATOR_JOB	No	No
Recruiting Administrator View All	RECRUITING_ADMINISTRATOR_VIEW_ALL_DATA	No	No
Recruiting Manager View All	RECRUITING_MANAGER_VIEW_ALL_DATA	No	No
Time and Labor Administrator View All	TIME_AND_LABOR_ADMINISTRATOR_VIEW_ALL_DATA	No	No
Time and Labor Manager View All	TIME_AND_LABOR_MANAGER_VIEW_ALL_DATA	No	No

The roles already assigned to the integration user; none were added for OAuth.

Part 2: Import the certificate as a trusted partner certificate

In the Fusion environment's identity domain, I switched the console to the domain's home region and opened Security > Trusted partner certificates. I imported the certificate generated in Cloud Shell, using the same alias as the signing key in Oracle Integration.



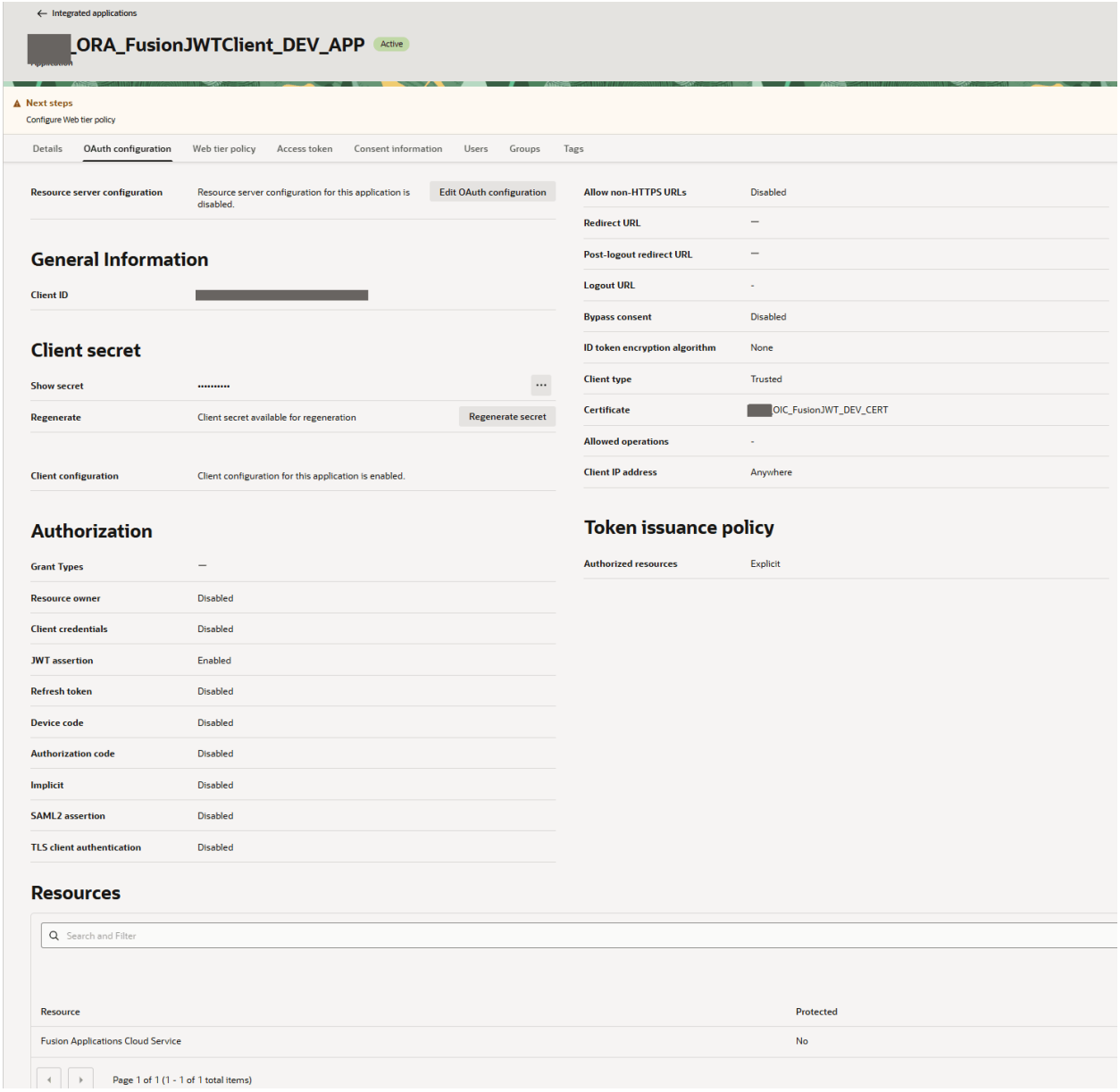
The certificate imported as a trusted partner certificate in the Fusion identity domain.

Part 3: Create the confidential application

Under Integrated applications, I added a confidential application and configured it as a client. I enabled only the JWT assertion grant type, set the client type to Trusted, and selected the imported certificate. I set Client IP address to Anywhere, because the form leaves it empty and will not save otherwise.

For the token issuance policy, I added the Fusion resource. In this domain it is listed as Fusion Applications Cloud Service. The resulting scope has two parts joined without a space: urn:opc:resource:faaas:fa:<POD>

followed by urn:opc:resource:consumer::all. The application also generates a client secret, which JWT User Assertion does not use, and shows a Web tier policy banner that can be ignored.



The confidential application: JWT assertion only, Trusted client with the certificate, and the Fusion resource.

Part 4: Prove the token from Cloud Shell

Before touching Oracle Integration, I wrote a short Python script in Cloud Shell. It read the private key from the vault, signed a user assertion and a client assertion with the same claims Oracle Integration uses, requested a token from the identity domain, and then called a Fusion HCM REST resource with that token.

The token request returned 200 with the integration user as the subject, and the Fusion call returned 200. The token expired in 300 seconds because its lifetime follows the assertion's exp claim; Oracle Integration

creates a new assertion for each request, so this does not affect connections. I shredded the temporary key file afterwards.

```
mraylo@cloudshell:    -oauth (us-phoenix-1)$
mraylo@cloudshell:    -oauth (us-phoenix-1)$ python3 token_test.py
Token request: 200
sub: SVC_OIC_NONPROD | expires in 300 s
Fusion GET: 200
mraylo@cloudshell:    -oauth (us-phoenix-1)$ shred -u key.pem && ls -l ~/    -oauth
total 8
-rw-------. 1 mraylo oci 1159 Oct 29 04:55     _OIC_FusionJWT_DEV_CERT.cer
-rw-------. 1 mraylo oci 1539 Oct 29 06:22 token_test.py
mraylo@cloudshell:    -oauth (us-phoenix-1)$
```

The token test: token request 200 for the integration user, and a Fusion REST GET returning 200.

What I learned

Creating everything in the Fusion environment's identity domain, and in that domain's home region, avoided the most common mistake. The resource name and the empty Client IP field were small surprises that cost time. Proving the token outside Oracle Integration first meant that, when I switched the connections, I only had to troubleshoot Oracle Integration and not the identity setup.